

Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)
Course Name: Applied Mathematics-II Course Code: MA-201(L) Examination- 2nd periodical
Scheme- CBCS Branch: CSE(AI/ML) Sem: Second
Question Paper for 2nd Periodical Examinations (May, 2024)

Time: 2 Hours

Max. Marks: 50

Note: All questions are compulsory.

Course Outcomes:

(Dr. Kiran Thakur)

On successful completion of this course the students should be able to:

CO1. Calculate rank, Eigen values, Eigen vectors, inverse of matrix by Jordan method and other matrix transformations.

CO2. Determine solution of system of linear equations, basis and dimension, rank-nullity theorem and analyze Cayley Hamilton Theorem.

CO3. Solve the differential equations using appropriate method and apply the concept in complex engineering problems.

CO4. Develop the concepts of Laplace transformation and inverse Laplace transform with its properties to solve initial and boundary value problems.

CO5. Illustrate and apply the Fourier series to find representation of different functions.

CO6. Discuss and apply vector calculus in different types of integral problems along with Green, Stokes and Gauss divergence theorem.

Q.1 (a) Define Fourier series of a function for the change of interval and write the formulae of the Fourier Coefficients.

(b) State Dirichlet Conditions for Fourier series.

(CO5)

(c) If the $L(f(t)) = f(s)$ then prove that $L(f(at)) = (1/a) f(s/a)$.

(d) Define unit step function?

(e) What is laplace transform? Discuss the conditions for its existence?

(5×2=10)

Q.2 (a) (CO4) Find the laplace transform of $(\sqrt{t} - 1/\sqrt{t})^3$.

(5)

(b) (CO4) Find the laplace transforms of the following function $(\sin 2t - \sin 3t)/t$.

(5)

Q.3 (a) (CO4) Find the inverse laplace transform of $s/(s^4+s^2+1)$.

(5)

(b) (CO4) Solve the differential equation by transform method

$d^2y/dt^2 + 2 dy/dt + 5y = e^{-t} \sin t$, where $y(0) = 0$ and $y'(0) = 1$.

(5)

Q.4 (a) (CO5) Expand $f(x) = ((\pi-x)/2)^2$ as a fourier series in $0 < x < 2\pi$.

(5)

(b) (CO5) Expand $f(x) = \pi x - x^2$ in a half range sine series in the interval $(0, \pi)$ up to first three terms.

(5)

Q.5 (a) (CO1) Solve the initial value problem by using laplace transform method,

$y'' + 7y' + 10y = 4e^{-3t}$, $y(0) = 0$ and $y'(0) = -1$.

(5)

(b) (CO5) Find the Fourier series to represent the function $|x|$, $-\pi < x < \pi$.

(5)